UNIVERSITY OF CALIFORNIA SAN DIEGO

Sensing Through Shape: Geometry-Encoded 3D Force Estimation for Continuum Robots

A thesis submitted in partial satisfaction of the
requirements for the degree Masters of Science

in

Engineering Sciences (Controls & Mechatronics)

by

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Committee in charge:

Professor Tania Morimoto, Chair
-
-

2025

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University of California San Diego

2025

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ABSTRACT OF THE THESIS

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by

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Masters of Science in Engineering Sciences (Controls & Mechatronics)

University of California San Diego, 2025

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Accurate force estimation in surgical continuum robots is critical for preventing tissue damage during minimally invasive procedures. Body-mounted force sensors offer direct measurements but are often bulky, difficult to sterilize, and impractical clinically. Base-mounted sensing is an appealing alternative, yet typically provides too few measurements to uniquely recover all contact variables—especially for 3D, multi-load interactions—and is highly susceptible to noise. In this work, we introduce a geometry-optimization framework that overcomes these

limitations by encoding missing information directly into the structure of the sensor. Inspired by the evolved diversity of animal whiskers, the method treats backbone geometry—such as cross-sectional shape, taper, and twist—as tunable design variables that embed positional or angular information when direct measurements are insufficient. These geometric encodings are integrated into a nonlinear least-squares estimator implemented in the SoRoSim modeling environment, enabling full 3D multi-load force estimation using only proximal measurements. Simulation and hardware experiments on a tendon-driven continuum robot prototype show that optimized geometries, combined with small active "whisking" perturbations, resolve the ambiguity inherent to base-mounted sensing and yield up to a 10× improvement in spatial acuity and an 8× increase in force-estimation accuracy compared to conventional cylindrical designs.

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If you want an even tighter, more high-impact conference-ready version, I can do that too.

Chapter 1

Introduction to Force Estimation

1.1 Current Force Estimation Methods

In this section we explore existing force estimation methods for tendon-driven continuum robots. Specifically we look at:

1) measurement types—what is being measured.

- Visual/Shape: Sense body shape/strain (e.g., cameras, markers, FBGs)
- Proximal-Only

2) Types of models—how measurements are used to estimate the applied force.

- Data driven/learning based
- Physics-based continuum mechanics model.
- FEM model

1.2 Whisker-Based Force Estimation in Animals

In this section we discuss how animals use whiskers to sense their surroundings.

- Whiskers act as biological cantilever beams
- Tactile information is encoded in the mechanical signals at the base.

- Animals often struggle to differentiate between transverse forces and bending moments.
- Whisker design varies by species
- Taper improves discriminability by preventing full fold-back.
- Animals actively move (whisk) their whiskers

1.3 Contributions

In this work, we introduce several contributions to the estimation of external forces in tendon-driven continuum robots.

- ***1) Geometry-Encoded Sensing Capabilities:*** We show that optimized geometries enable accurate estimation of complex loading scenarios—including multiple simultaneous forces and 3D contacts with limited sensor channels—that conventional cylindrical backbones cannot resolve due to inherent sensing degeneracies.
- ***2) Geometry-Optimization Framework:*** We introduce a general framework that, given the available proximal measurements and desired contact variables, automatically identifies backbone geometries that minimize estimation error under noise. The method optimizes radius profiles, taper, twist, and cross-sectional shape to encode information that is otherwise unobservable.
- ***3) Hardware Implementation and Validation:*** We fabricate these optimized geometries in a 3D-printed tendon-driven continuum robot and validate performance using a custom rig with closed-loop winches for precise, repeatable 3D force application, overcoming the limitations of manual loading.

Chapter 2

Whisker-Based Sensing Approach

2.1 Non-uniqueness of the inverse mapping

We now examine conditions under which the mapping between an applied external load $F = [F, s, \theta_1]$ and the resulting proximal reaction wrench W is unique. If each load F produces a wrench W that differs (within tolerance) from the wrench generated by any other load, then the forward and inverse mappings are both unique, and F can be recovered from measurements of W. Conversely, if two distinct loads produce wrenches that are indistinguishable within tolerance, the inverse problem is non-unique, since multiple loads correspond to the same measured wrench [1].

1) Proximal Small-Force Degeneracy: Consider the simple Euler-Bernoulli beam subject to a point force of magnitude F at position s. The deflection at the tip is:

$$\delta(L) = \frac{F(3Ls^2 - s^3)}{6EI} \tag{2.1}$$

and the sensitivity functions are:

$$\frac{\partial F}{\partial \delta} = \frac{6EI}{(3Ls^2 - s^)3} \qquad \frac{\partial s}{\partial \delta} = \frac{EI}{F(Ls - \frac{1}{2}s^2)} \tag{2.2}$$

Equation (2.2) shows that force estimation F depends only on position s. As $s \to 0$ (proximal contact), $\frac{\partial F}{\partial \delta} \to \infty$, so small deflection noise produces large force error. Therefore,

force estimates are most accurate distally and degrade proximally. Position estimation s depends on both s and F. As $F \to 0$, $\frac{\partial s}{\partial \delta} \to \infty$, giving high sensitivity and poor position accuracy for small forces. As $s \to \infty$, $\frac{\partial s}{\partial \delta} \to \infty$, again yielding high sensitivity and poor accuracy for very proximal contacts. Overall, small proximal forces are the hardest to estimate, while large distal forces are the easiest. We also see increasing stiffness, with E or I makes both estimates worse. Therefore, a softer beam creates larger deflection per unit force, higher SNR, and generally a better estimation.

2) Available Measurements: The uniqueness of the inverse mapping depends on the available proximal measurements. With access to the full 6D proximal reaction wrench $W = [T_x, T_y, T_z, F_x, F_y, F_z]$, a single applied point force is uniquely recoverable in the noise-free case.However, when the sensing is restricted—for example, to only three channels $[T_y, T_z, F_x]$-the applied force is not guaranteed to be uniquely determined [2]. Thus, choosing a smaller or cheaper sensor introduces a fundamental tradeoff: reduced sensor bulk comes at the cost of reduced information and potential non-uniqueness. Rats overcome this sensing tradeoff with a simple geometric solution: tapered whiskers. Because transverse forces and bending moments are difficult to distinguish at the follicle, taper makes the whisker itself a more informative mechanical sensor, improving spatial acuity without adding new sensory channels. Prior work [2] shows that the mapping from the proximal triplet $[T_y, T_z, F_x]$ to radial contact distance is always unique for a tapered whisker, but non-unique for a cylindrical one. This result is further supported by [3], which shows that cylindrical whiskers produce overlapping $[T_y, F_x]$ pairs for different applied forces, leading to non-uniqueness, whereas tapered whiskers generate clearly separated pairs (Fig. 2.1).

3) Large Deflection Degeneracy: At large deflections, a cylindrical whisker can fold back on itself, creating non-unique base signals (Fig. 2.1). In this regime, a highly deflected distal contact can produce the same reaction forces as a moderately deflected proximal one. This fold-back occurs only in cylinders because their uniform stiffness allows the beam to enter a constant-curvature (circular-arc) regime, whereas tapering increases distal flexibility and prevents

this behavior. In this work very large deflections are not reached and mostly ignored. However, for more torturous paths this degeneracy should be considered.

2.2 Robustness to Noise

In this section we discuss how the tapered whisker while having excellent performance is not robust to noise and why.

- Discuss how animals use array's of whiskers to overcome noise.
- Introduce how animals use not only absolute base measurements at the whisker follicle (noise prone), but relative measurements (noise resistant), and changes in measurements over time to estimate velocity.
- Discuss how sensing is more resistant to noise by the whisker tip and sensitive to noise by the whisker base. (why animals have multiple whiskers of different lengths).

Under zero noise, the tapered design provides clearly distinguishable base measurements, and, therefore, a unique mapping from base measurements to applied force (Fig. 2.3). Under realistic noise, however, this advantage collapses, and the tapered design often performs no better, than the cylindrical design (Fig. 2.4). Therefore, while the tapered design improves performance over the cylindrical design, it lacks robustness to noise. Rats use several strategies to overcome noise and maintain accurate spatial awareness.

1) Multiple Whiskers of Varying Lengths: Their differently sized whiskers provide simultaneous, redundant measurements that average out noise and support reliable distance estimation at both proximal and distal contact points. Equation.(2.2) tells us that small proximal forces are the most sensitive to noise, having many whiskers with varied lengths and stiffness helps compensate.

2) Absolute, Relative, and Time-Based Sensing: Rats also rely on relative rather than absolute follicle signals: by taking multiple measurements from a single whisker in slightly

different configurations, they cancel out systematic error and resist additive noise. Changes in follicle signals over time also let rats estimate velocity.

3) Passive vs. Active Sensing:

- Discuss how animals brush their whisker over an object to obtain a richer proximal signal.
- Discuss absolute, difference-based, and time based sensing methods.

4) Weber's Law: Finally, their whisker sensory processing follows Weber's Law, making small changes easier to detect when the overall stimulus is weak [3]. By making the rat more sensitive to small mechanical signals, Weber's law ensures that smaller bending moments and forces still produce high resolution estimates.

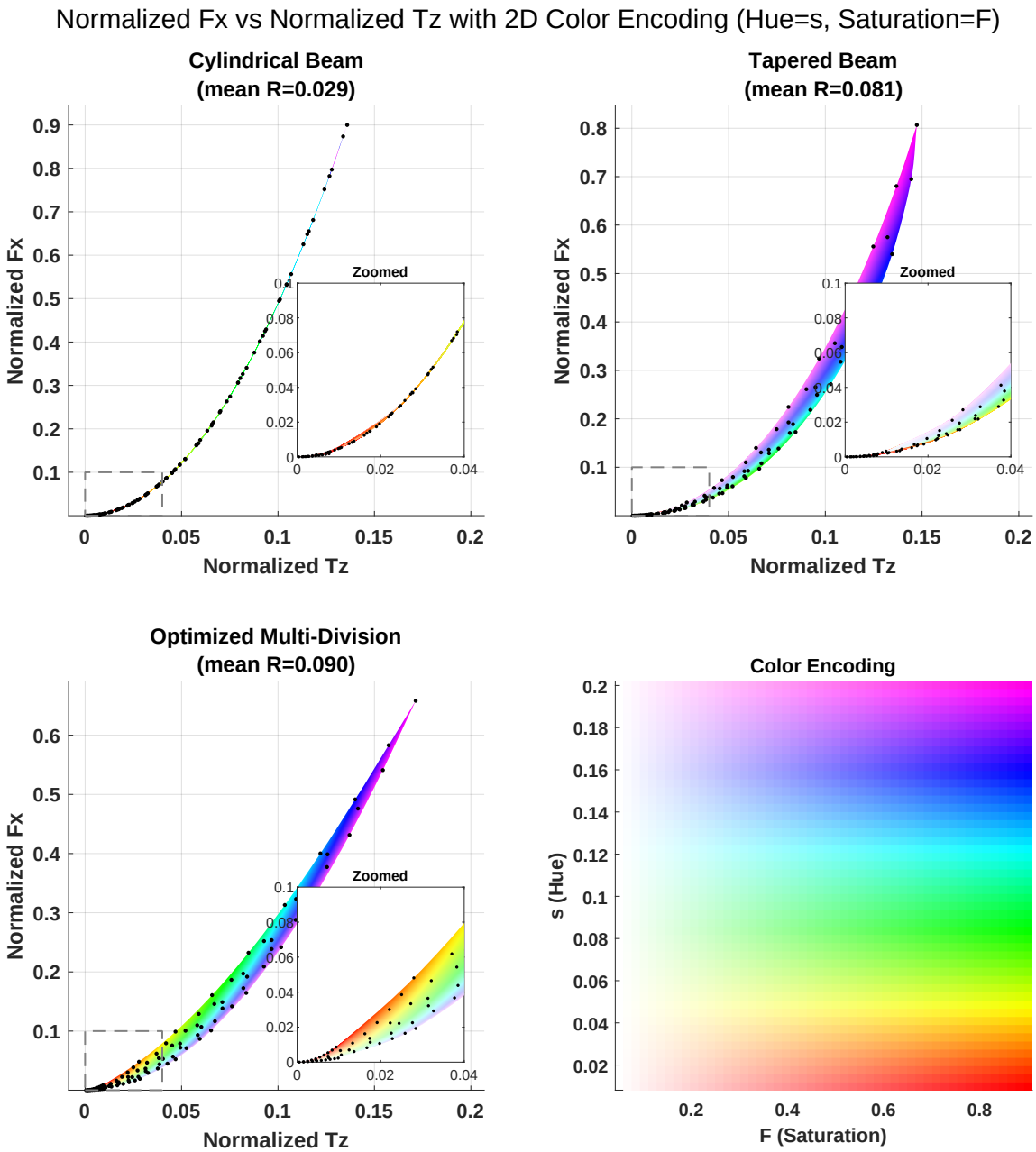


Figure 2.1. Plot of axial force F_x versus bending moment T_y at the base, computed for a range of applied external loads for two geometries: a cylindrical design ($r = 1.5\,\text{mm}$) and a tapered design ($r = [1.5\,\text{mm}, 1.0\,\text{mm}]$). The results show that, for the tapered geometry, F_x and T_y vary independently, enabling unique identification of the applied external load from the measured base reactions. The cylindrical geometry produces many overlapping (F_x, T_y) pairs, making the applied loads non-unique and therefore not uniquely identifiable. The plot illustrates how spread out the (Fx, Ty) points are. Greater separation among these points indicates a more unique mapping, making it less likely that noise in (Fx, Ty) will propagate into error when solving for

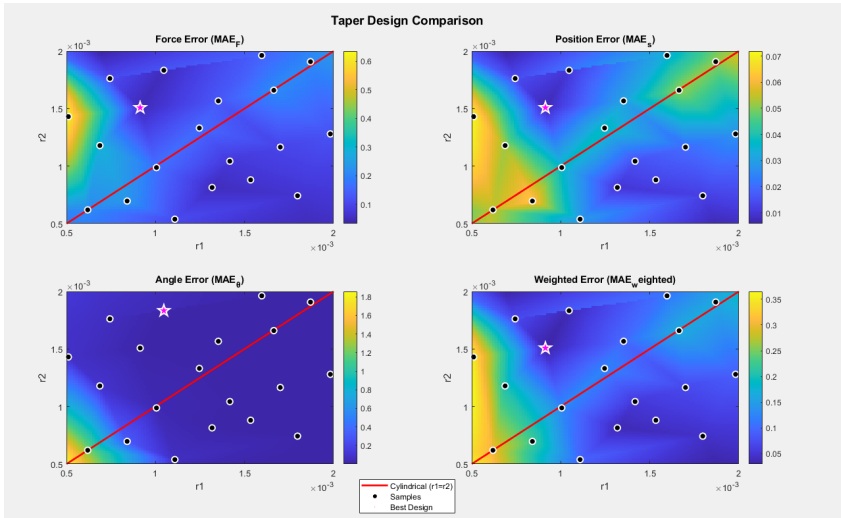
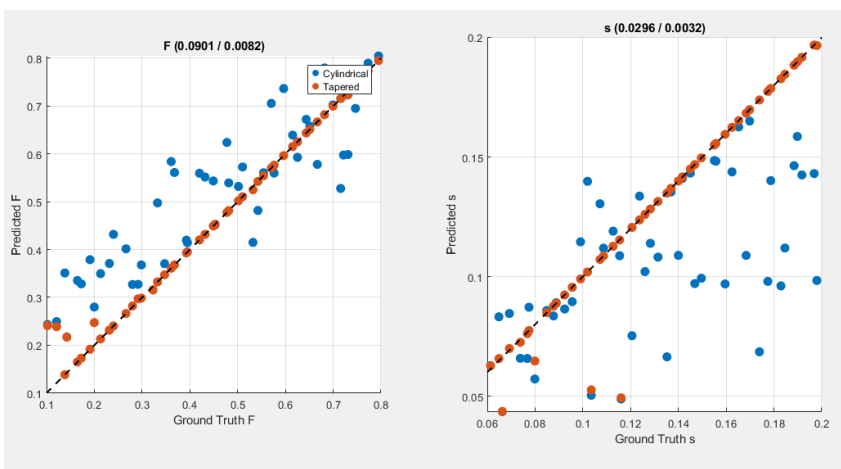


Figure 2.2. Heatmap illustrating how different combinations of proximal radius r_1 and distal radius r_2 influence the accuracy of the estimated external load $\hat{x} = [\hat{F}, \hat{s}, \hat{\theta}_1]$



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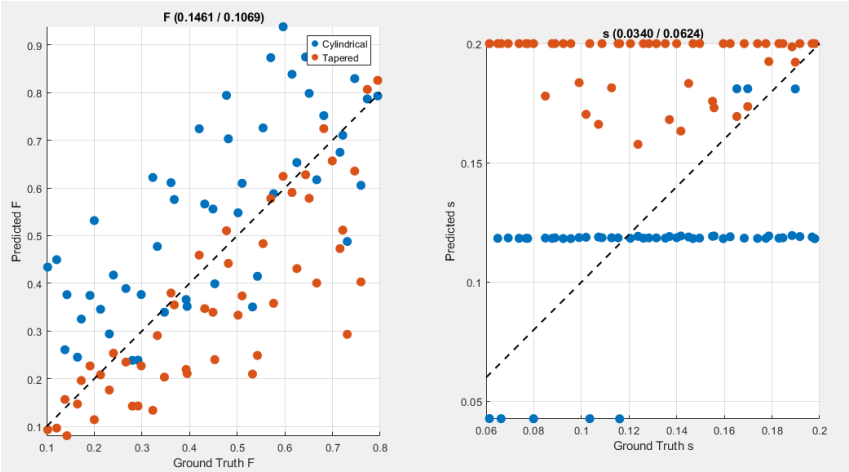

Figure 2.4. assuming additive noise with std dev: $[\sigma T_y, \sigma T_z, \sigma F_x] = [0.005Nm, 0.005Nm, 0.07N]$

Chapter 3

Method

In this chapter we explore the methodology used in this work.

3.1 SoRoSim Modeling Framework

The robot statics are modeled using SoRoSim, a MATLAB-based toolbox implementing the Geometric Variable-Strain (GVS) formulation of Cosserat rod theory for continuum robots. This framework was selected for its computational efficiency and native support for tendon-driven actuation.

3.1.1 Model Configuration

We model our robot as a fixed cantilever beam initially orientated along the x-axis. The distal end is unconstrained with three degrees of freedom, twist about the x-axis, bending about the y-axis, and bending around the z-axis. The proximal end is fixed to the global coordinate frame origin.

To quantify reaction forces and moments at the base, a six-degree-of-freedom constraint (three translational, three rotational) is enforced at the proximal end. This constraint is position-controlled, fixed at a specified position and orientation by the user. SoRoSim then allows us to solve the six-dimensional wrench required to maintain this specified fixed configuration.

To create different backbone designs with different cross-sectional areas, we utilized attributes of the SorosimLink class. This class allows the user to create a rigid or soft body which

can later be attached via joints to other links to form a linkage. Link divisions allow you to split the link into discretized intervals, each interval with its own properties.

3.1.2 Tendon Actuation Model

Three tendons are routed along straight, constant-radius paths at 120° circumferential intervals with an 8 mm radial offset from the backbone centerline. Each tendon path is modeled as a constraint curve rather than through explicit routing disk geometry, a technique demonstrated in [1] and [2]. This simplification assumes the physical routing disks are sufficiently thin that their bending stiffness contribution is negligible compared to the backbone, reducing model complexity while maintaining computational efficiency.

Each tendon is independently actuated via tension control $[\tau_1, \tau_2, \tau_3]$. Tensioning a tendon pulls the robot backbone toward the sensor, generating a wrench. Since we aim to measure only wrenches from external forces, we needed to decouple wrenches caused by tendon actuation alone. To achieve this, we mounted the tendon actuation units proximal to the 6-axis force-torque sensor. This design ensures that tendon forces remain internal to the system, preventing the sensor from directly measuring wrenches due to tendon actuation in isolation. However, the tendons still indirectly affect measurements: by pulling a tendon, the backbone's equilibrium configuration changes, which alters the mapping between applied external loads and measured base reactions.

3.1.3 Applied Point Force Model

While calculating the proximal wrench from an applied force is straightforward, the inverse problem is often ill-conditioned. Multiple distinct point forces can produce identical or nearly identical proximal wrenches, making them difficult to distinguish—particularly when sensor noise and resolution limits further reduce distinguishability [3]. This ill-conditioning also degrades the convergence speed of force estimates.

Parameterizing applied loads as Gaussian distributions helps improve conditioning. Stud-

ies have shown this approach improves both convergence speed and accuracy while enhancing robustness to measurement noise [3]. We therefore implement Gaussian load parameterization in this work.

Applied forces are characterized by three components:

- F: force magnitude
- s: position along the total arc length relative to the origin
- θ: angle about x-axis $[0, 2\pi]$

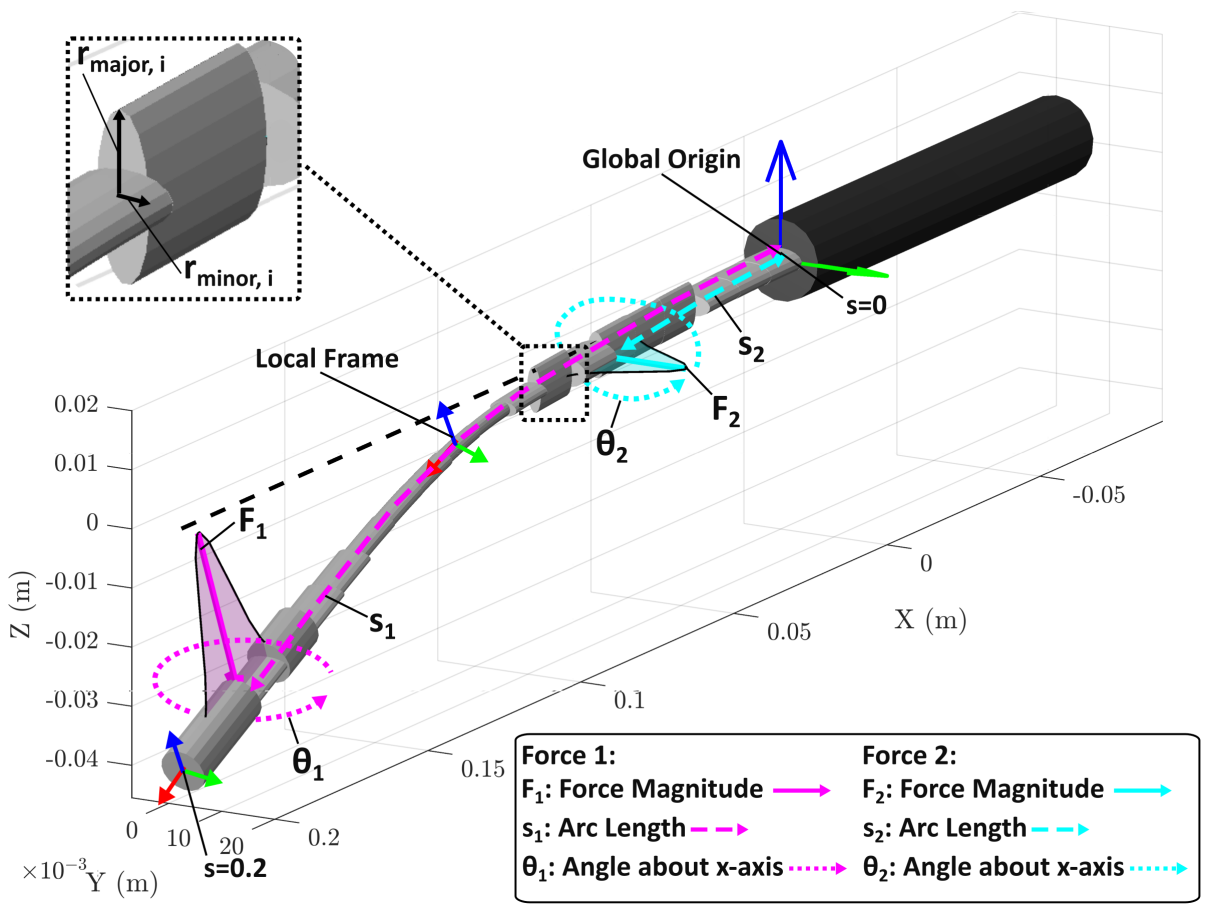

Figure 3.1. Schematic showing: a) force application to the robot, b) position of the 6 DOF virtual joint, c) global and local coordinate frames, d) force parameterization as a discrete Gaussian distribution.

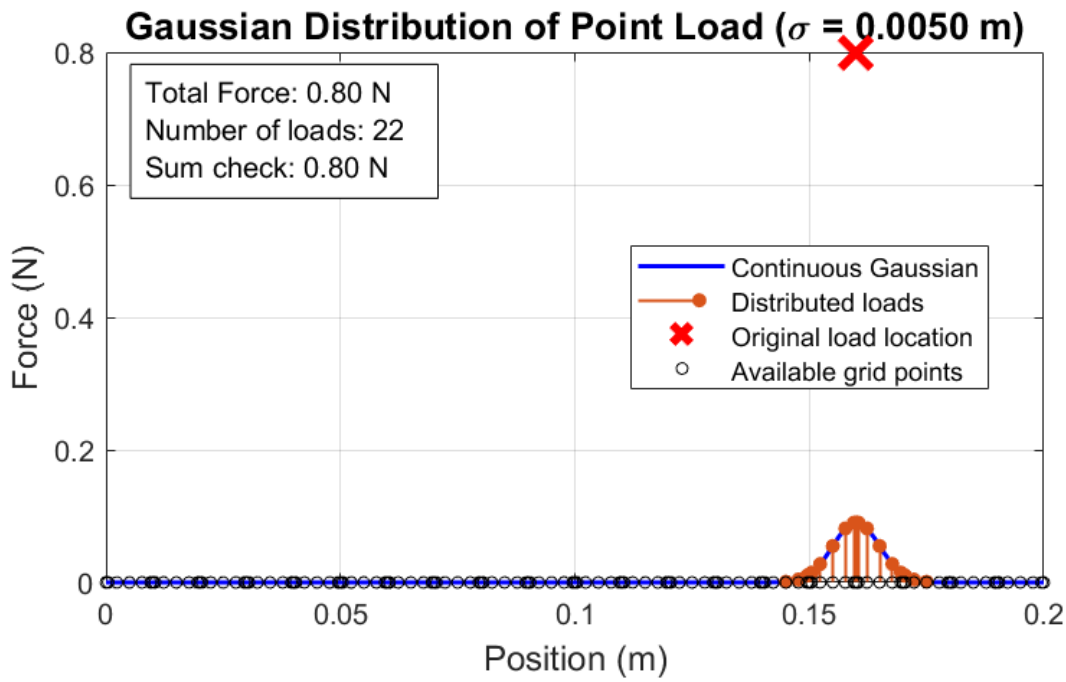

Figure 3.2. Illustration of parameterizing a point force as a discretized Gaussian distribution with a standard deviation of $\sigma = 0.005$ by default. Replacing the point load with a narrow Gaussian improves numerical conditioning and reduces computation time in the inverse solver. The original point force at position s is represented by a set of smaller forces whose magnitudes sum to the original load and follow the prescribed Gaussian profile.

The applied forces are assumed to be transverse, acting perpendicular to the longitudinal axis, and concentric, passing through the geometric center of the backbone. They are also modeled as follower forces, meaning their orientation is defined relative to the local frame at each application point rather than the global reference frame. Follower forces were chosen to facilitate visualization and reduce computational time for static equilibrium calculations.

3.1.4 Forward Model

The forward model is computed using an author-created MATLAB function:

$$W(x,\tau) = forward_model(x,\tau) \in \mathbf{R}^6 \tag{3.1}$$

- **Input:** Applied Force $x = [F, s, \theta]$ and tendon tension $\tau = [\tau_1, \tau_2, \tau_3]$

- **Output:** Resulting proximal wrench $W = [T_x, T_y, T_z, F_x, F_y, F_z]$
- **Method:** SoRoSim $statics()$ function
- **Computation time:** Approximately $0.05s$
- **Assumptions:**
  - **Transverse Forces:** Applied perpendicular to the local longitudinal axis.
  - **Concentric Forces:** Pass through the geometric center (no offset).
  - **Follower Forces:** Defined relative to the local frame, not the global frame.
  - **Elastic Deformation:** Purely elastic (non-plastic) with full recovery.
  - **Static Equilibrium:** Strictly time-invariant; no kinetic motion or vibration.
- **Multiple point forces:** For N applied point forces, stack vertically $x = \begin{bmatrix} F_1, s_1, \theta_1 \\ \vdots \\ F_N, s_N, \theta_{1N} \end{bmatrix}$

The function takes a user-defined input force and passes it to a custom function that computes the corresponding discretized Gaussian force distribution and applies it to the robot. It then runs SoRoSim's statics function to perform a static equilibrium simulation. SoRoSim outputs the proximal wrench required to hold this quasi-static equilibrium configuration under the applied load as well as the deflected backbone shape. An optional plotting function can then visualize the deflected backbone, tendons, and applied forces.

3.1.5 Inverse Model

The inverse model for a single point force is computed using an author-created MATLAB function:

$$\hat{x} = [\hat{F}, \hat{s}, \hat{\theta}] = inverse_model(W_m, \tau_m) \tag{3.2}$$

- **Input:** Measured proximal wrench $W_m = [T_x, T_y, T_z, F_x, F_y, F_z]$ (via ATI Mini40 IP65 6-axis force-torque sensor) and measured tendon tensions $\tau = [\tau_1, \tau_2, \tau_3]$ (via load cells)

- **Output:** Estimated external applied force $\hat{x} = [\hat{F}, \hat{s}, \hat{\theta}]$

- **Multiple point forces:** To decide if the system is experiencing one or two forces, the model tests both scenarios against the proximal wrench data. It applies the Akaike Information Criterion (AIC) to identify the best fit, ensuring the chosen model is accurate without being unnecessarily complex. For N applied point forces, solution is stacked vertically $\hat{x} = \begin{bmatrix} \hat{F}_1, \hat{s}_1, \hat{\theta}_1 \\ \vdots \\ \hat{F}_N, \hat{s}_N, \hat{\theta}_N \end{bmatrix}$

- **Objective:** The solver attempts to minimize the the weighted difference between the measured wrench W_m and the simulated wrench W produced by the forward model.

- **Residual Weighting:** Residuals are weighted to reduce the influence of noisy measurements and prioritize reliable ones, preventing high-variance channels from dominating the estimation.

  - **Noise-Based Scaling:** Each output wrench component is scaled by the inverse of its noise standard deviation, σ_{noise}.

  - **Statistical Trust:** High-noise channels are strongly downweighted and only influence the solution when their signal is large, while low-noise channels dominate the estimation.

- **Method:**

  - **Global Initialization (LHS):** Latin Hypercube Sampling (LHS) generates 100 candidate solutions. The five with the lowest residual norms are selected as initial guesses x_0.

  - **Local Refinement (`lsqnonlin`):** Each x_0 is refined using MATLAB's trust-region-reflective nonlinear least-squares solver until an optimality tolerance of 10^{-10} is reached. The solution with the lowest final residual norm is selected.

- **Constraints:** Lower and upper bounds $lb = [0.1, 0.01, 0]$ and $ub = [1, 0.2, 2\pi]$ constrain the design variables.

3.2 Design Optimization

3.2.1 Overview

This design optimization seeks to determine the backbone geometry of a tendon-driven continuum robot. The goal is to enable the accurate and concise estimation of two external point loads using only a single sensor.

System Description

The backbone transmits mechanical signals to the proximal end, where a 6-axis force-torque sensor measures the resulting wrench, $W_m = [T_x, T_y, T_z, F_x, F_y, F_z]$. This measurement is used to estimate the properties of the two contact forces, defined as $\hat{x} = [\hat{F}_1, \hat{s}_1, \hat{\theta}_1, \hat{F}_2, \hat{s}_s, \hat{\theta}_2]$.

The Problem: Cylindrical Backbones

Preliminary testing indicates that a standard cylindrical backbone cannot provide an accurate estimation. Physically, when a concentric load is applied to a cylinder, it bends but does not twist, resulting in a zero torsion measurement ($T_x = 0$).

Mathematically, this creates an underdetermined problem. You are attempting to estimate 6 unknown force parameters using only 5 effective measured wrench parameters. Analyzing the force-to-wrench Jacobian (J) reveals that the matrix is rank-deficient:

$$J(\mathbf{x}) = \frac{\partial \mathbf{W}}{\partial \mathbf{x}} \tag{3.3}$$

$$\text{rank}(J) < 6 \quad \text{(for a cylindrical backbone under concentric loading)} \tag{3.4}$$

The Solution: Elliptical Geometry

To guarantee a unique solution, we require a backbone geometry where $\text{rank}(J) = 6$. Furthermore, the system must be well-conditioned to prevent sensor noise from causing large estimation errors.

This is achieved by using an elliptical backbone. Unlike a cylinder, an ellipse lacks rotational symmetry. Therefore, concentric loads induce a twisting moment. By dividing the beam into discrete elliptical segments, we can vary the dimensions along the length. This creates a unique stiffness pattern, coupling force to torsion differently at every position, which allows the sensor to localize the applied load.

3.2.2 Optimization Objective

The primary objective is to maximize the observability of all force components. We achieve this by maximizing the minimum singular value, σ_{min}, of the Jacobian matrix.

- A positive σ_{min} guarantees the matrix is full rank.
- A larger σ_{min} improves conditioning. Since $1/\sigma_{\text{min}}$ acts as the "gain" for error amplification, maximizing σ_{min} minimizes the worst-case error derived from sensor noise.

Robustness via Sampling

To ensure the sensor works for a wide range of scenarios, we use Latin Hypercube Sampling (LHS) to select a well-distributed set of test forces. For each configuration, we compute a normalized Jacobian (J_{norm}) using the following steps:

1. **Perturbation:** We use central finite differences, perturbing inputs (F, s, θ) by $\pm\varepsilon$.
2. **Input Scaling:** Inputs are normalized by their physical range (e.g., 1.0 corresponds to full effort).
3. **Output Scaling:** Outputs are normalized by the sensor noise floor σ_{sensor}, converting them to a Signal-to-Noise Ratio (SNR).

4. **Evaluation:** We apply Singular Value Decomposition (SVD) to J_{norm}. The smallest singular value represents the worst-case SNR.

We define the residual vector as the inverse of these singular values (minimizing the inverse is equivalent to maximizing the value):

$$r_i = \frac{1}{\sigma_{\min,i}} \tag{3.5}$$

3.2.3 Design Variables

The backbone is discretized into $N = 20$ segments along its length. This number was chosen as it provides a good balance between geometric customization and ease of manufacturing. The optimization algorithm modifies the geometric parameters of each individual segment to find an optimal design. The design vector, $\mathbf{x}$, is a concatenation of the geometric properties for all 20 segments. For an elliptical cross-section, these properties include the major axis diameter (d_{maj}), the minor axis diameter (d_{min}), and the orientation angle (ψ) of the ellipse.

Therefore, for $N = 20$ segments, the optimization vector is defined as:

$$\mathbf{x} = \begin{bmatrix} d_{\text{maj},1} & d_{\text{min},1} & \psi_1 & \cdots & d_{\text{maj},20} & d_{\text{min},20} & \psi_{20} \end{bmatrix}^T \tag{3.6}$$

The optimization is constrained by physical manufacturing limits and material properties, defined by lower bounds ($\mathbf{l}$) and upper bounds ($\mathbf{u}$):

$$\mathbf{l} \leq \mathbf{x} \leq \mathbf{u} \tag{3.7}$$

3.2.4 Optimization Strategy

The problem is formulated as a weighted non-linear least squares optimization. We employ a two-stage approach to mitigate local minima:

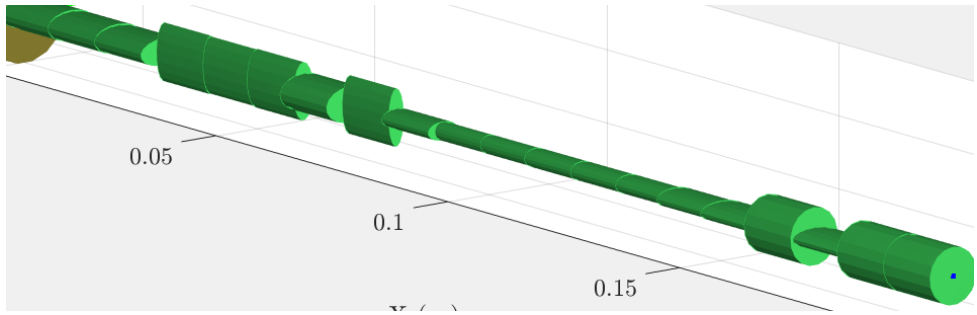

Figure 3.3. Schematic showing an example of an optimal design with $N = 20$ elliptical segments.

1. **Global Initialization (LHS):**
   - We generate 500 random design candidates spanning the parameter space using Latin Hypercube Sampling.
   - The candidate with the lowest residual is selected as the initial guess, x_0.
2. **Local Refinement (`lsqnonlin`):**
   - We refine x_0 using the Trust-Region-Reflective algorithm.
   - Gradients are computed via forward finite differences.
   - The solver iterates until optimality tolerances (10^{-10}) or function tolerances (10^{-8}) are met.

The final mathematical formulation is:

$$\begin{aligned} \underset{\text{Design } \mathbf{x}}{\text{minimize}} \quad & \sum_{k=1}^{N} \left(\frac{1}{\sigma_{\min}(J(\mathbf{x}, \text{load}_k))} \right)^2 && \leftarrow \text{Minimize worst-case errors} \\ \text{subject to} \quad & \mathbf{l} \leq \mathbf{x} \leq \mathbf{u} && \leftarrow \text{Stay within physical limits} \end{aligned} \tag{3.8}$$

3.3 Fabrication

3.3.1 Robot Fabrication

We selected a 200*mm* backbone length to balance adequate reach with manufacturability on standard FDM 3D printer beds. We chose Bambu's PLA Basic as our printing material with the

following material properties referenced from their datasheet (Young's modulus $E = 2.58GPa$, Poisson's ratio $\nu = 0.34$). Robot designs were prototyped in SolidWorks and printed on the Bambu X1 Carbon for a precise physical model.

3.3.2 Test Rig Fabrication

To compare the force estimation performance of different geometric designs, we built a motorized testing setup to apply two point forces and apply three tendon tensions.

Three winches positioned at 120° intervals around the circumference of the designated beam enable independent closed-loop tension control of each tendon. Each winch uses a NEMA 17 stepper motor and TMC2209 stepper motor driver to wind the tendon in and out.

The tendon wraps around a pulley configuration that directs it through a load cell. When tensioned, the tendon pulls a center piece into the load cell, which measures a force component proportional to the total tension (the proportion depends on the pulley angles). The system was calibrated using known masses.

The load cell provides high-bandwidth feedback to an Arduino Mega microcontroller running a proportional control algorithm that reduces motor speed as tension approaches the target value.

The adjustable frame provides three degrees of freedom: two rotations and translational adjustment of the force application point by repositioning the hook attachment along the beam's length. An additional winch mounted on the upper frame applies a target external point force. Motor tension targets are set via the Arduino IDE serial interface.

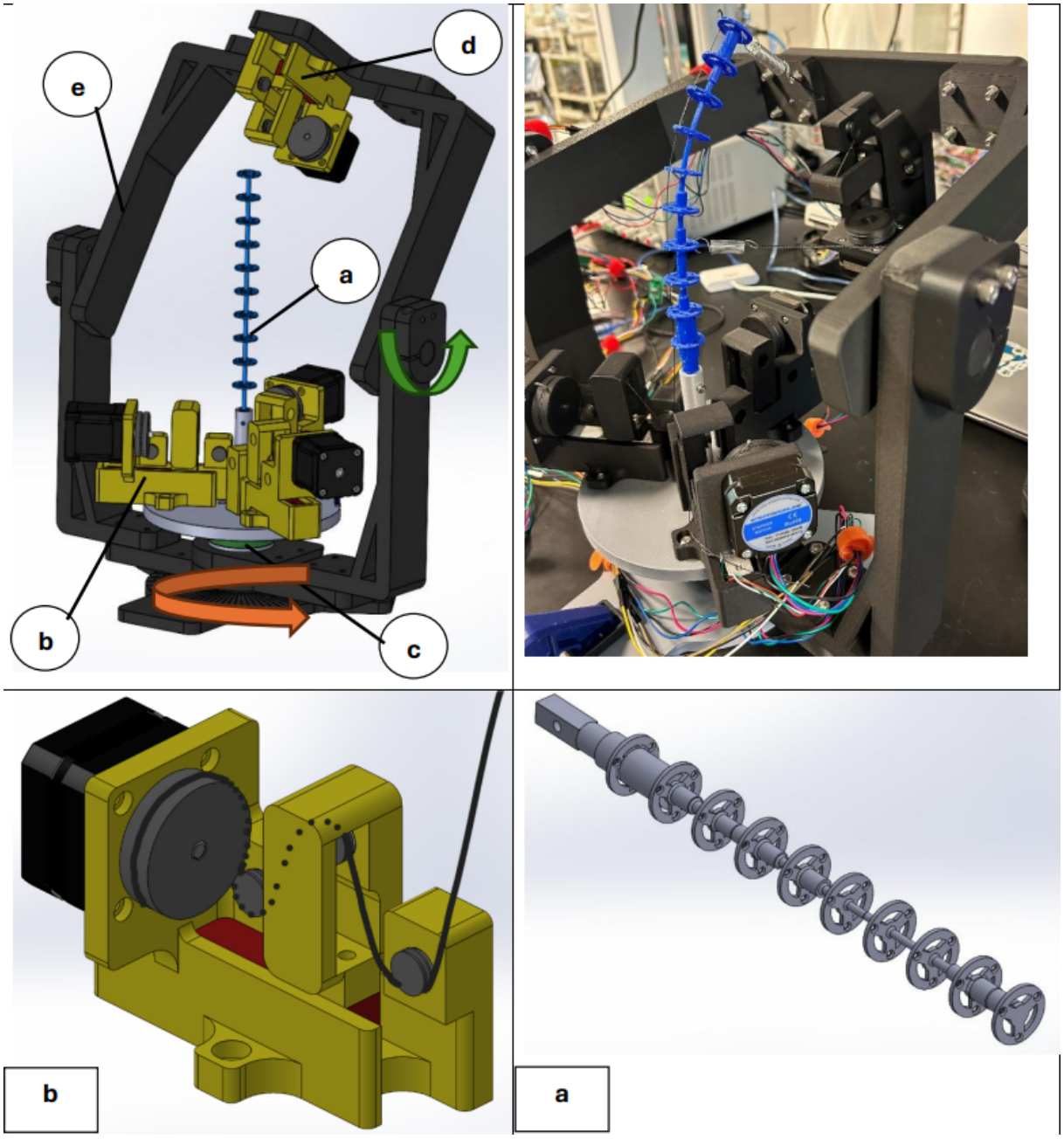

Figure 3.4. Schematic showing: (a) the main continuum robot body; (b) a closed-loop tension-control winch for reaching and maintaining a desired tendon tension; (c) a swivel adjustment frame for setting θ_1; (d) a closed-loop tension-control winch for applying and holding an external point force; and (e) a pitch adjustment frame for setting θ_2.

Chapter 4

Experimental Validation

In this chapter we experimentally validate our model and hypothesis that changing the backbone geometry can improve accuracy of force estimation.

4.1 SoRoSim Model Validation

In this section, we compare the theoretically predicted proximal wrenches with those obtained experimentally across a range of applied forces, thereby demonstrating the effectiveness of SoRoSim as our modeling framework.

Plot: Scatter Plot: SoRoSim Predictions vs. Experimental Ground Truth: 6 subplots one per Dependent Variable $[T_x, T_y, T_z, F_x, F_y, F_z]$ with error bars

4.2 Robotic Setup

4.3 Procedure

In this section we discuss: the design of experiment, the force combinations we tested, how we conducted the experiment, how we ensure fairness across designs and trials.

For hardware validation we tested two designs:

4.4 Data Analysis

In this section we discuss: How the raw data was processed.

Design	Length (mm)	Proximal Radius (mm)	Distal Radius (mm)
Cylindrical	200	1.5	1.5
Tapered	200	1.7	1.25
Optimized	200	2	1

Table 4.1. Tested Designs

Factor Name	# of factor levels	Levels
F_1	10	0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0
s_1	5	$90°, 135°, 180°, 225°, 270°$
θ_2	1	$90°$

Table 4.2. Experimental Trials

4.5 Results

In this section we present the results through a series of plots.

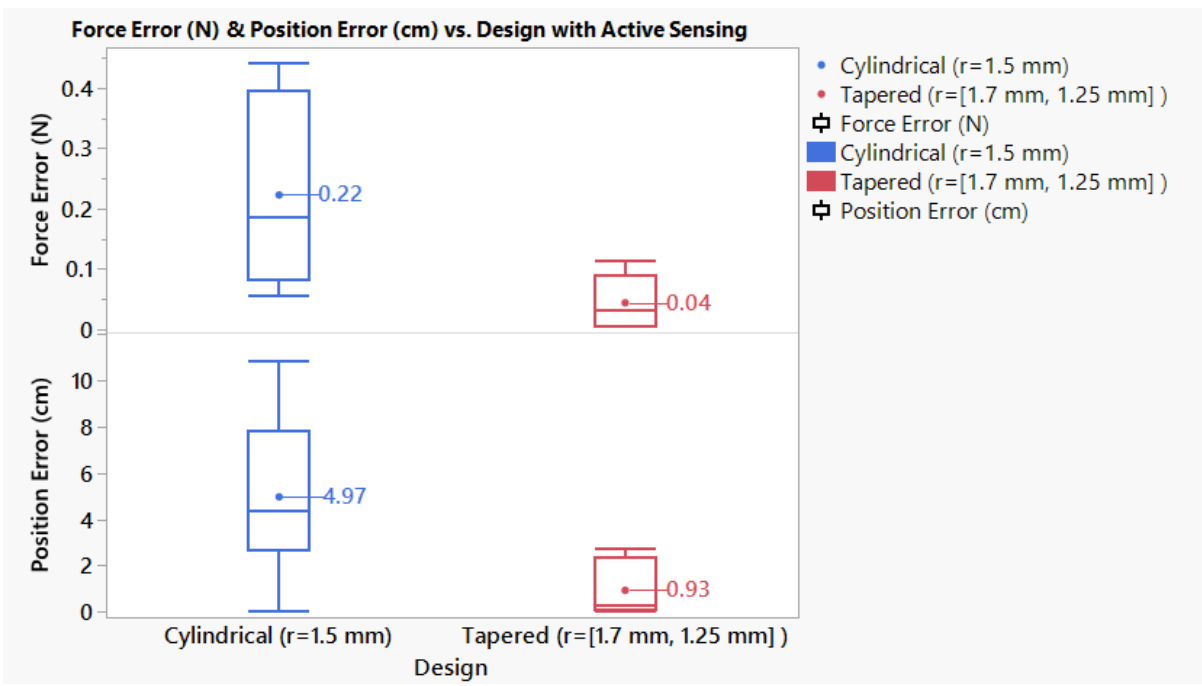

Figure 4.1. Box plot comparing the absolute estimation error in $\hat{x} = [\hat{F}, \hat{s}, \hat{\theta}_1]$ obtained from measured experimental data for two geometries: a cylindrical design ($r = 1.5\,\text{mm}$) and a tapered design ($r = [1.5\,\text{mm}, 1.0\,\text{mm}]$), using active sensing. The plot highlights the improvement in estimation accuracy and precision achieved with the tapered geometry.

Plot 3—Heatmap: 4 Heatmaps showing (a) force error, (b) position error, (c) angle error,

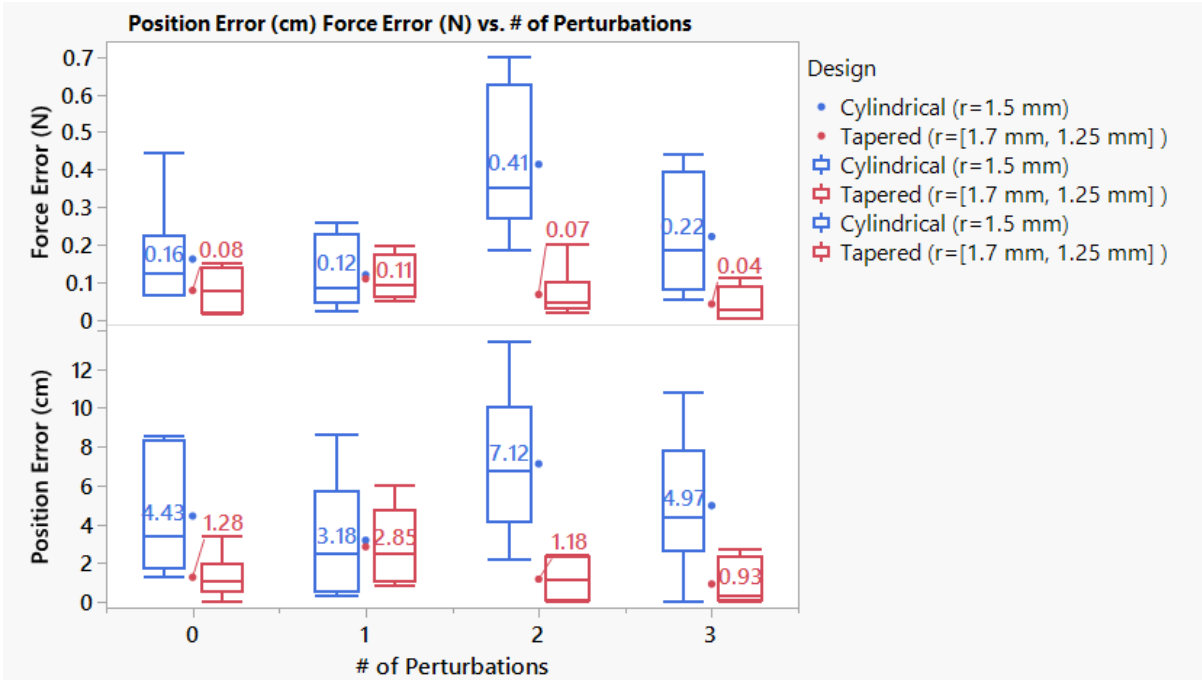

Figure 4.2. Box plots illustrating the effect of the number of whisks on the absolute estimation error in $\hat{x} = [\hat{F}, \hat{s}, \hat{\theta}_1]$ computed from experimental measurements for two geometries: a cylindrical design ($r = 1.5\,\text{mm}$) and a tapered design ($r = [1.5\,\text{mm}, 1.0\,\text{mm}]$). The results show consistently lower estimation errors for the tapered geometry, with further reductions observed as the number of whisks increases.

and (d) combined error. x and y axis are s and F respectively.

Chapter 5

Discussion

Chapter 6

Conclusion and Future Work

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